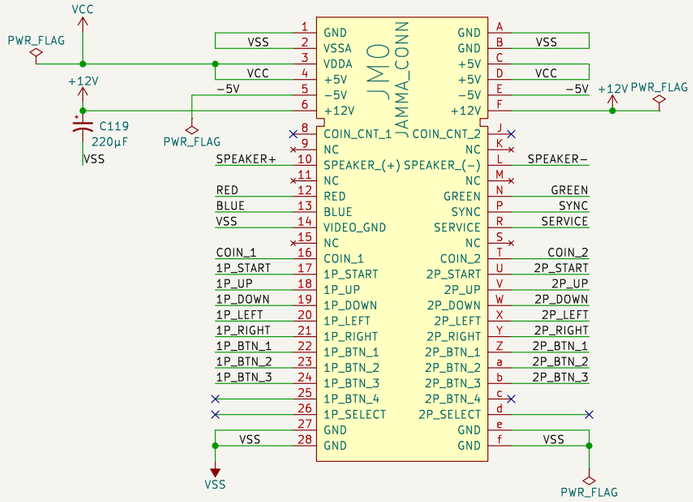
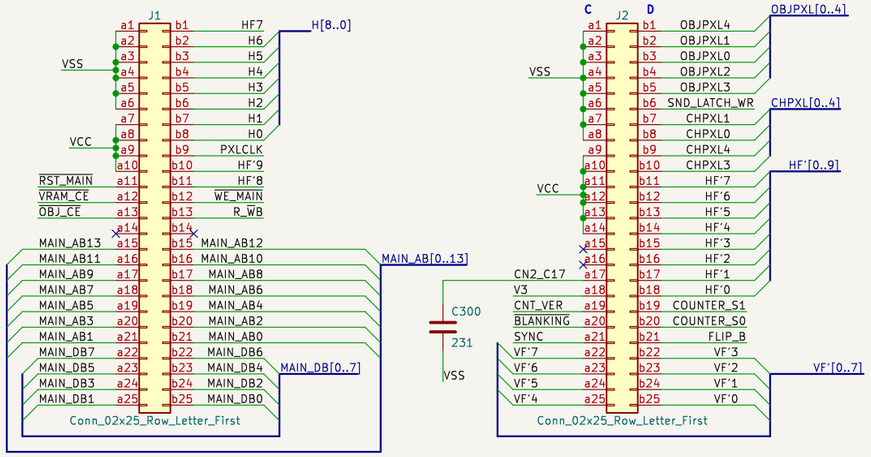
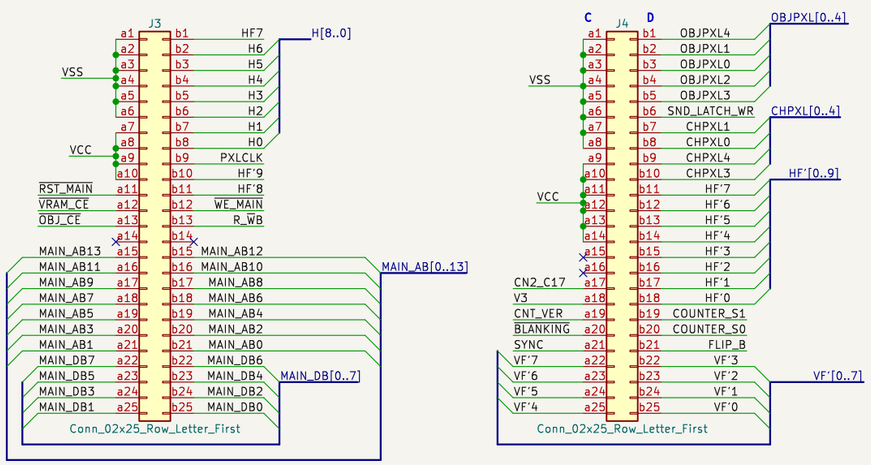
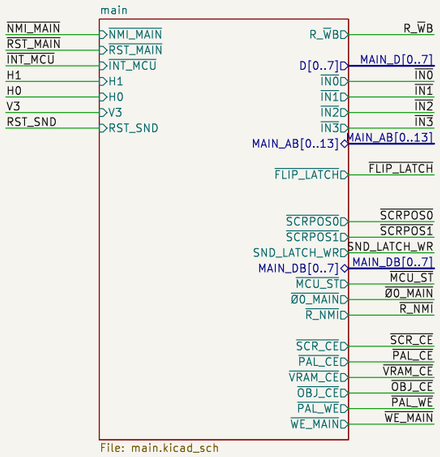


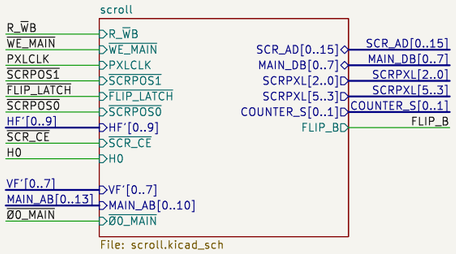
Top board connectors

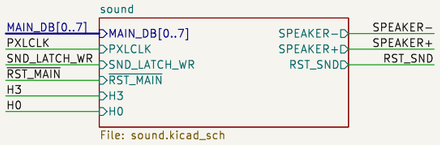


Bottom board connectors



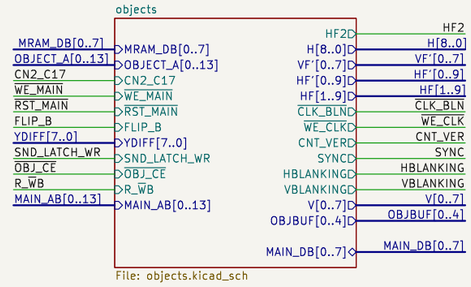


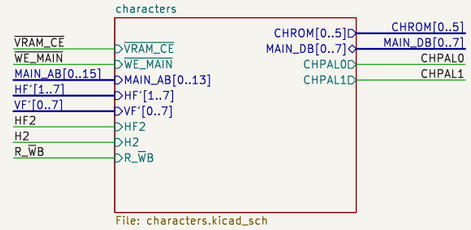


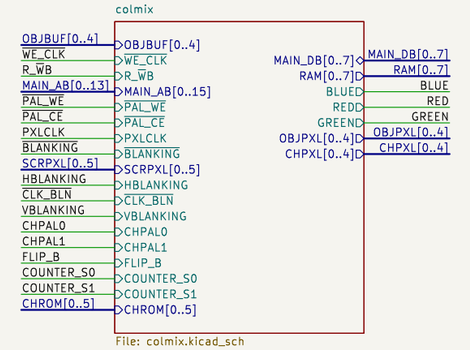


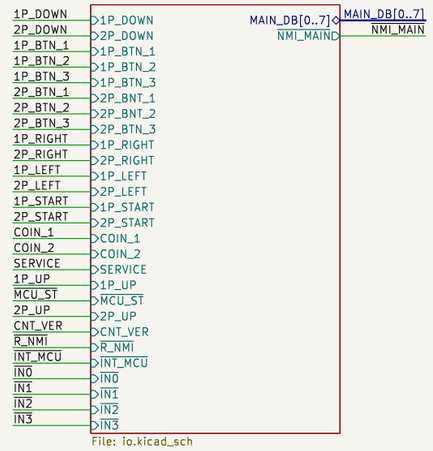












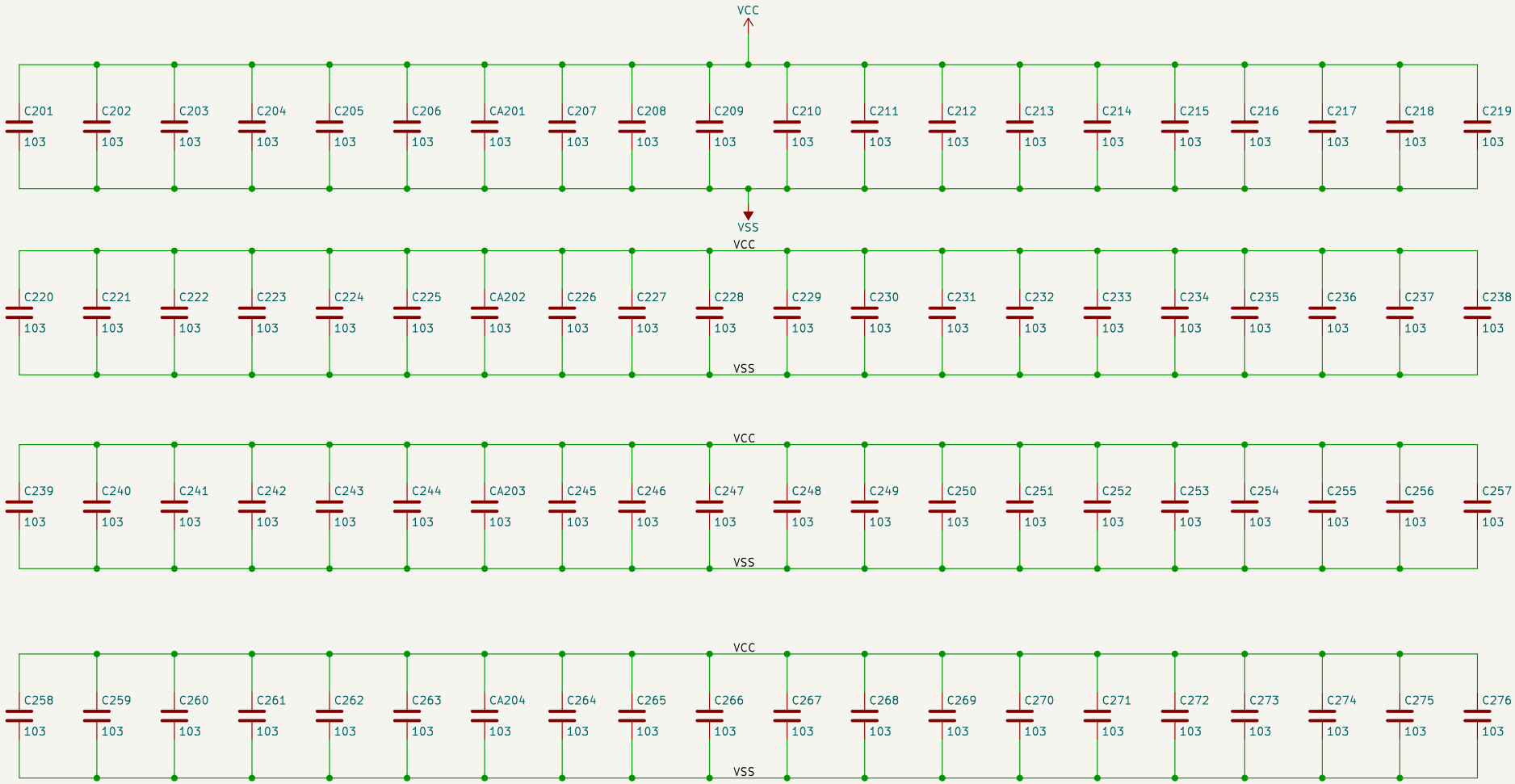
For repair and maintenance

JOTEGO

Sheet: /
File: kunio.kicad_sch

Title: RENEGADE

Size: A3	Date: 2022-07-08	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 1/16



For repair and maintenance

JOTEGO

Sheet: /capacitors/
File: capacitors.kicad_sch

Title: RENEGADE

Size: A4	Date: 2022-07-08	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 2/16

SND_D[0..7]

MAIN_DB[0..7]

SND_A[0..14]

A[14..0]

D[7..0]

VCC

VSS

RESET

R49 4.7kΩ

R41 1kΩ

R42 47Ω

C96 4.7µF

F15D 74LS14

F15C 74LS14

RST_SND

A6 3-state flip flops 74LS374

SND_LATCH_WR

SND_LATCH_RD

74LS74

B5 MC68A09

R47 30KΩ

FIRQ

IRQ

NMI

RW_SND

E_SND

DMA/BREQ

HALT

MRDY

XTAL

EXTAL

PXLCLK

B1 HM65256B_SERIES 32kB RAM

A1 27256

D6A 74LS04

74LS00 A5B

E7C 74LS32

D6B 74LS04

F2 YM3812-3526 FM OPL

3.58MHz

Y3 YM3014

G3B L8410-LM358N

G3A L8410-LM358N

C97 4.7µF

10Ω R46

74LS138 AAA4 address decoder

A4A 74LS08

A4C 74LS08

RAM_CE

SND_LATCH_RD

PCM_CTRL

PCM_START

PCM_STOP

RST_MAIN

WE

M9012-358N

L8410-LM358N

R36

R37

C100 100µF

+12V

H1 MB3730

BYPASS

FILTER

IN

OUT_M

OUT_I

INV. AMP

SPEAKER+

SPEAKER−

C102 4.7µF

C103 220µF

C101 4.7µF

NOT POPULATED

C99 10µF

R44 2.2kΩ

R45 4.7kΩ

RV1 R_Pot_Trim_US

C98 4.7µF

D1A 74LS393

D1B 74LS393

D3A 74LS393

D3B 74LS393

C1 27256

C3 27256

C4 27256

D4B 74LS74

74LS157 S Quad Mux

E2 OKI-M5205 Semiconductor converter 32kHz

DAOUT

VCK

XT

R43 1Ω

C104 103

resistor values likely to be wrong

M9012-358N

H3A

R48 1Ω

RN3 R_Network08_US

H3

PCM_RST

For repair and maintenance

JOTEGO

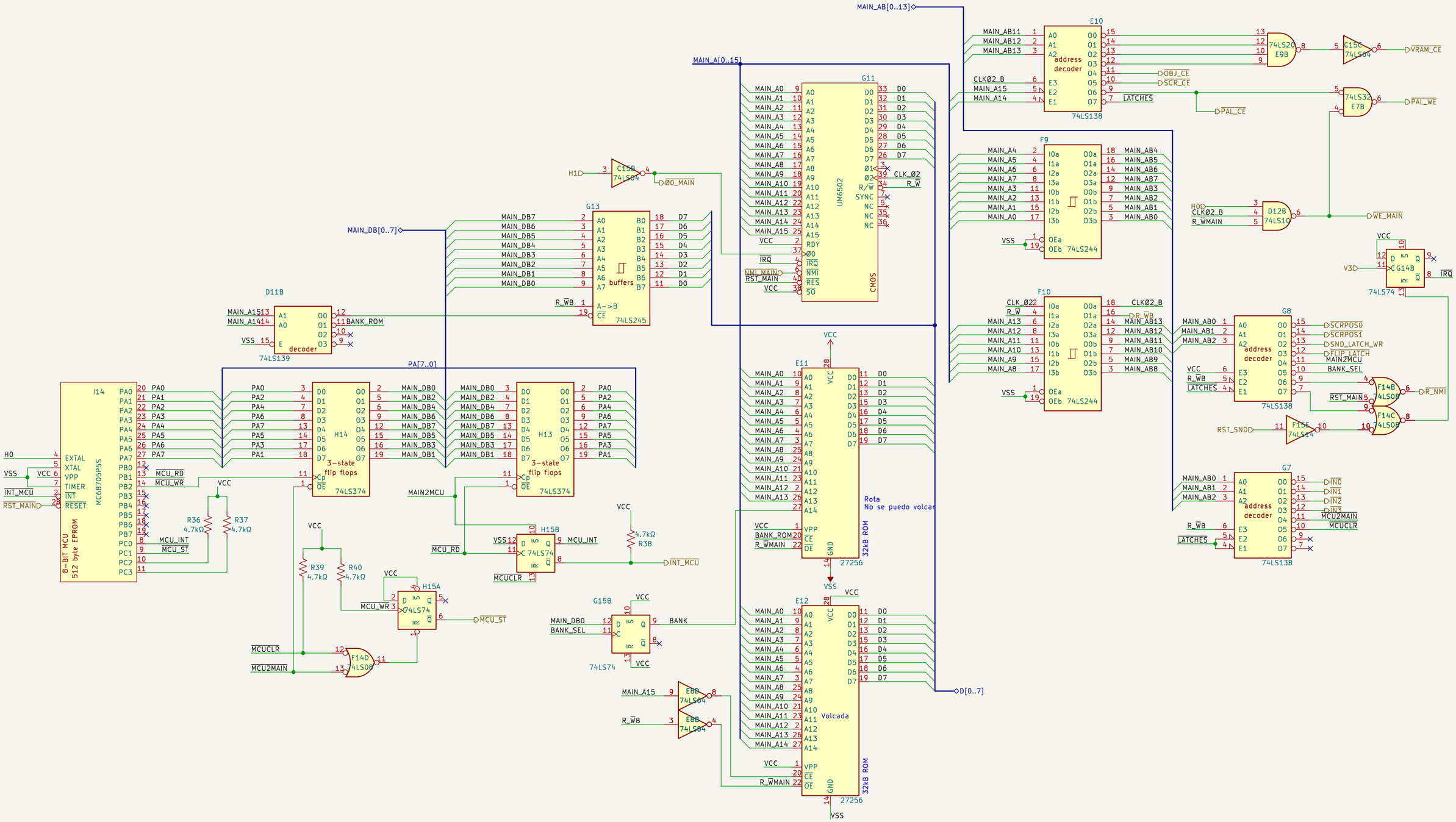
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File: sound.kicad_sch

Title: RENEGADE

Size: A3 | Date: 2022-07-08 | Rev: José Tejada

KiCad E.D.A. 8.0.9 | Id: 3/16



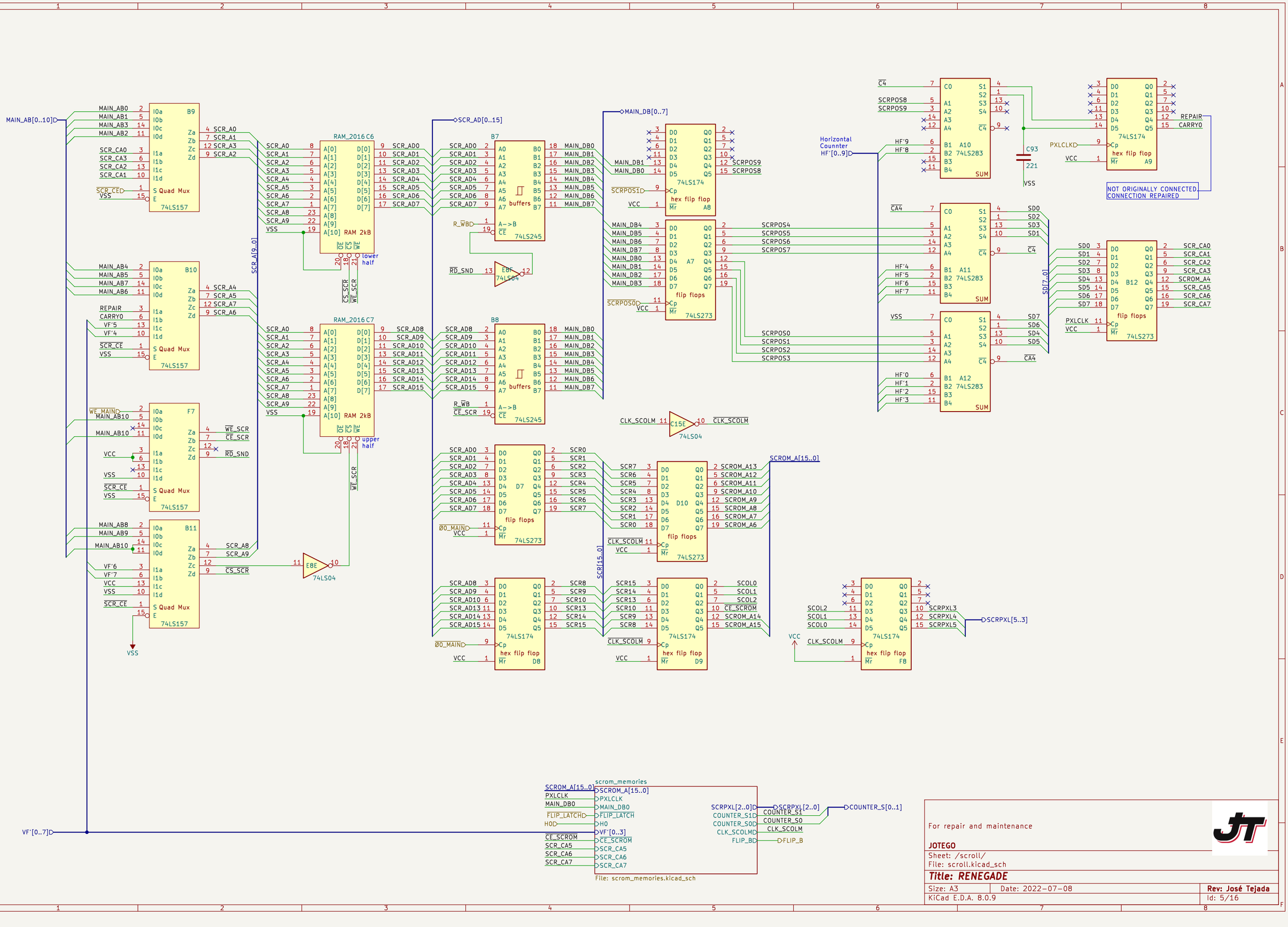
For repair and maintenance

JOTEGO

Sheet: /main/
File: main.kicad_sch

Title: RENEGADE

Size: A3	Date: 2022-07-08	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 4/16


MAIN_AB[0..10]
SCR_AD[0..15]
MAIN_DB[0..7]
SCR_A[9..0]
RAM_2016 C6
RAM_2016 C7
lower half
upper half
74LS157
74LS245
74LS174
74LS273
74LS04
buffers
hex flip flop
flip flops
Horizontal Counnter HF'[0..9]
NOT ORIGINALLY CONNECTED
CONNECTION REPAIRED
SCROM_A[15..0]
SCRPXL[5..3]
SD[7..0]
VF'[0..7]
scrom_memories
File: scrom_memories.kicad_sch
For repair and maintenance
JOTEGO
Sheet: /scroll/
File: scroll.kicad_sch
Title: RENEGADE
Size: A3
Date: 2022-07-08
Rev: José Tejada
KiCad E.D.A. 8.0.9
Id: 5/16

SCROM_A[15..0]

SCR_CA5 → C15F 74LS04 → SCROM_A5

CE_SCROM → C15A 74LS04 → CE_SCRB

D11A 74LS139 decoder
- CE_SCROM, SCROM_A14, VSS
- O0 CE_SCRF, O1 CE_SCRD, O2 CE_SCRE, O3 CE_SCRC

VF'[0..3]

A15 27256 — VCC, CE_SCROM, OE, GND, VPP
- VF'0–VF'3
- SCROM_A4–SCROM_A14
- D0 SCROM_D0, D1 SCROM_D1, D2 SCROM_D2, D3 SCROM_D3, D4 SCROM_D4, D5 SCROM_D5, D6 SCROM_D6, D7 SCROM_D7

B15 27256 — VCC, CE_SCRB, OE, GND, VPP
- VF'0–VF'3
- SCROM_A4–SCROM_A14
- D0 SCROM_D0, D1 SCROM_D1, D2 SCROM_D2, D3 SCROM_D3, D4 SCROM_D4, D5 SCROM_D5, D6 SCROM_D6, D7 SCROM_D7

SCROM_D[0..15]

74LS157 Quad Mux
- I0a SCROM_D0, I0b SCROM_D1, I0c SCROM_D2, I0d SCROM_D3
- I1a SCROM_D4, I1b SCROM_D5, I1c SCROM_D7, I1d SCROM_D6
- S SCROM_A15, E VSS
- Za 4, Zb 7, Zc 12, Zd 9

C14 27256 — VCC, CE_SCRC, OE, GND, VPP
- VF'0–VF'3
- SCROM_A4–SCROM_A15
- D0 SCROM_D8, D1 SCROM_D9, D2 SCROM_D10, D3 SCROM_D11, D4 SCROM_D12, D5 SCROM_D13, D6 SCROM_D14, D7 SCROM_D15

C11 27256 — VCC, CE_SCRD, OE, GND, VPP
- VF'0–VF'3
- SCROM_A4–SCROM_A15
- D0 SCROM_D8, D1 SCROM_D9, D2 SCROM_D10, D3 SCROM_D11, D4 SCROM_D12, D5 SCROM_D13, D6 SCROM_D14, D7 SCROM_D15

C12 27256 — VCC, CE_SCRE, OE, GND, VPP
- VF'0–VF'3
- SCROM_A4–SCROM_A15
- D0 SCROM_D8, D1 SCROM_D9, D2 SCROM_D10, D3 SCROM_D11, D4 SCROM_D12, D5 SCROM_D13, D6 SCROM_D14, D7 SCROM_D15

C9 27256 — VCC, CE_SCRF, OE, GND, VPP
- VF'0–VF'3
- SCROM_A4–SCROM_A15
- D0 SCROM_D8, D1 SCROM_D9, D2 SCROM_D10, D3 SCROM_D11, D4 SCROM_D12, D5 SCROM_D13, D6 SCROM_D14, D7 SCROM_D15

SR0-1, SR0-0

74LS194 shift register
- P3 SCROM_D15, P2 SCROM_D14, P1 SCROM_D13, P0 SCROM_D12 (E14)
- P3 SCROM_D11, P2 SCROM_D10, P1 SCROM_D9, P0 SCROM_D8 (D14)
- P3 P2 P1 P0 B13
- Cp PXLCLK, Cl VCC
- Dsl VSS, Q0 15, Q1 14, Q2 13, Q3 12, Dsr VSS

74LS157 Quad Mux
- I0a E15, I0b, I0c, I0d
- I1a, I1b, I1c, I1d
- S FLIP, E VSS
- Za SCRPXL1, Zb SCRPXL0, Zc SCRPXL2, Zd 9
- SCRPXL[2..0]

SCR_CA6 → D13B 74LS86 → F14A 74LS08 ← H0
SCR_CA7 → D13A 74LS86 → F14A 74LS08

G15A 74LS74
- D ← F14A 74LS08, C ← MAIN_DB0, PR ← FLIP_LATCH, CL ← VCC
- Q → 74LS157 I0a D15

74LS157 Quad Mux
- I0a D15, I0b, I0c, I0d
- I1a, I1b, I1c, I1d
- S FLIP, E VSS
- Za COUNTER_S0, Zb SR0-0, Zc COUNTER_S1, Zd SR0-1

FLIP → D13D 74LS86 ← SCROM_A4 → CLK_SCOLM

FLIP → D0E 74LS04 → FLIP_B

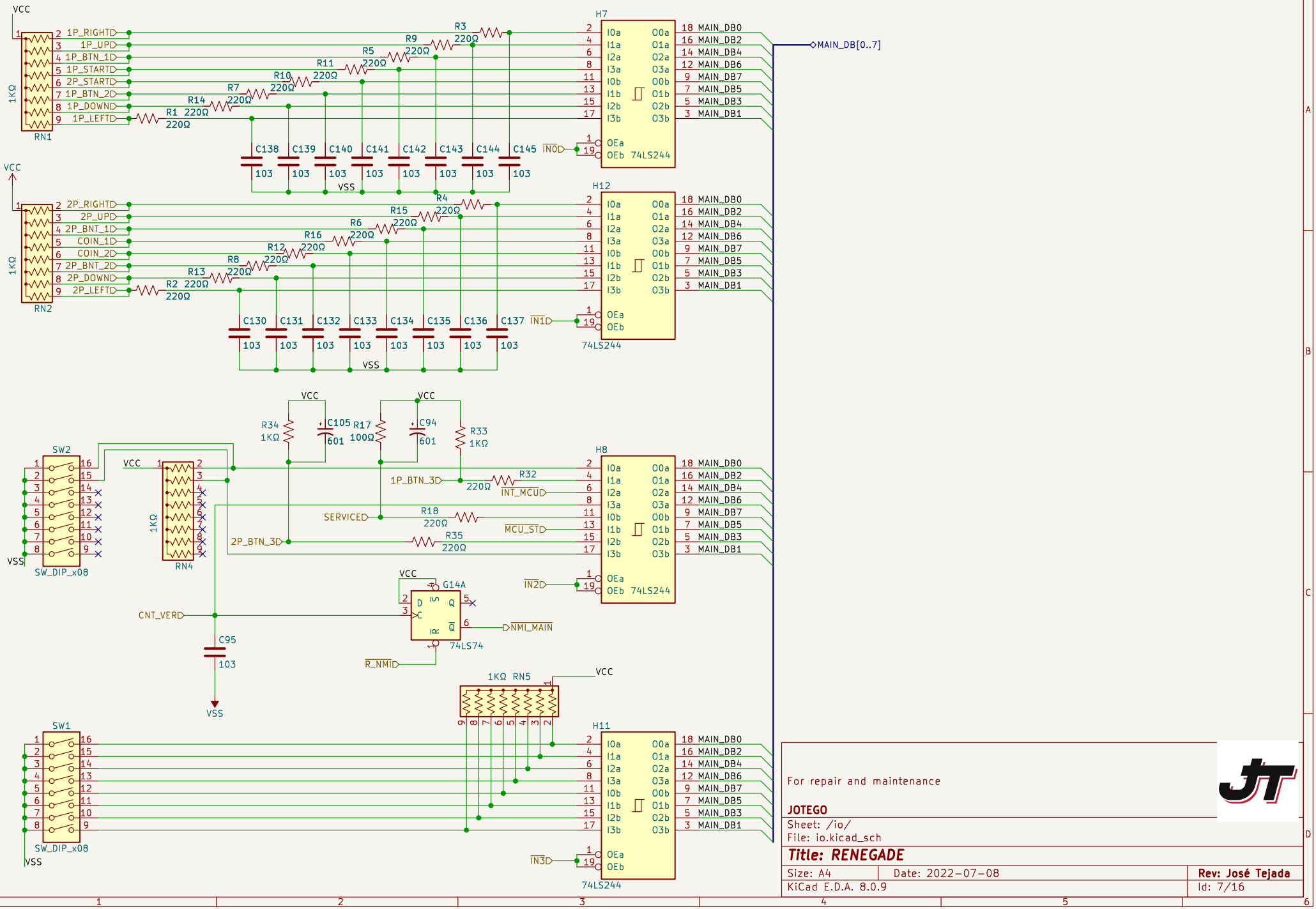
For repair and maintenance

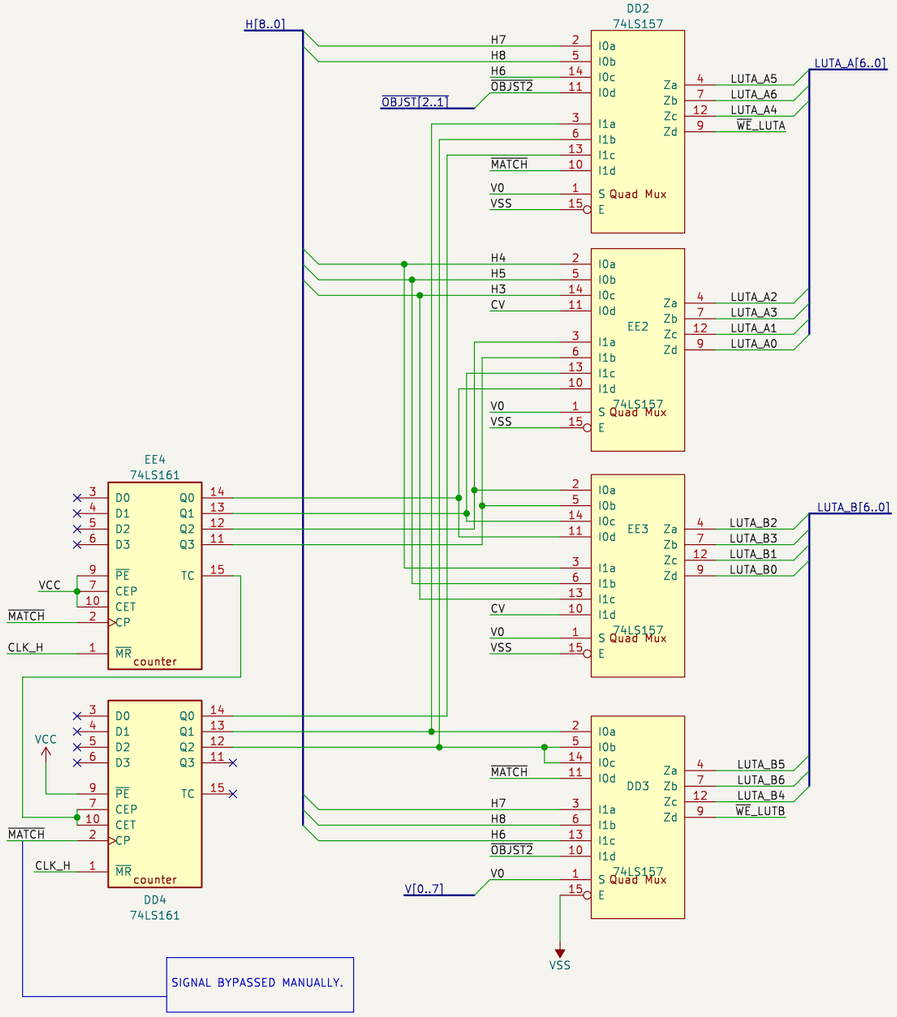
JOTEGO

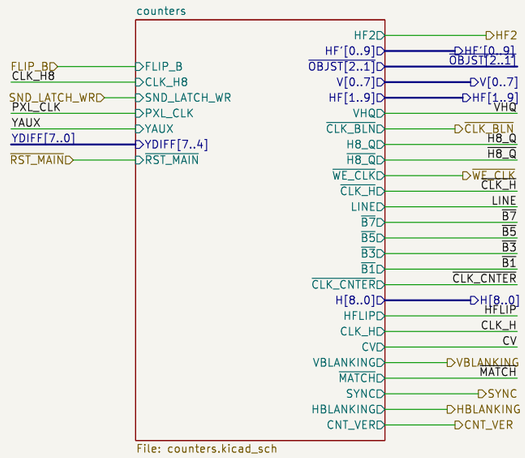
Sheet: /scroll/scrom_memories/
File: scrom_memories.kicad_sch

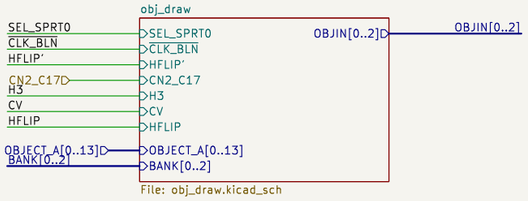
Title: RENEGADE

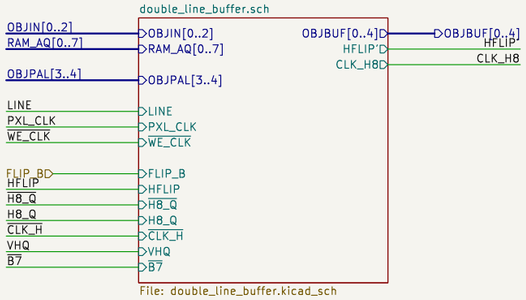
Size: A3	Date: 2022-07-08	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 6/16

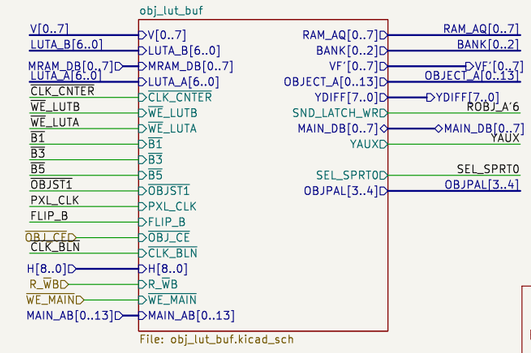

VCC
1P_RIGHT
1P_UP
1P_BTN_1D
1P_START
2P_START
1P_BTN_2D
1P_DOWN
1P_LEFT
RN1
1KΩ
R1 220Ω
R2 220Ω
R3 220Ω
R4 220Ω
R5 220Ω
R6 220Ω
R7 220Ω
R8 220Ω
R9 220Ω
R10 220Ω
R11 220Ω
R12 220Ω
R13 220Ω
R14 220Ω
R15 220Ω
R16 220Ω
C138 C139 C140 C141 C142 C143 C144 C145
C130 C131 C132 C133 C134 C135 C136 C137
103
VSS
H7
H12
H8
H11
74LS244
IN0D
IN1D
IN2D
IN3D
MAIN_DB[0..7]
MAIN_DB0
MAIN_DB1
MAIN_DB2
MAIN_DB3
MAIN_DB4
MAIN_DB5
MAIN_DB6
MAIN_DB7
2P_RIGHT
2P_UP
2P_BNT_1D
COIN_1D
COIN_2D
2P_BNT_2D
2P_DOWN
2P_LEFT
RN2
SW2
SW1
SW_DIP_x08
RN4
RN5
R34 1KΩ
R33 1KΩ
R17 100Ω
C105 601
C94 601
R32 220Ω
R18 220Ω
R35 220Ω
1P_BTN_3D
2P_BTN_3D
SERVICED
INT_MCU
MCU_ST
CNT_VER
C95 103
G14A
74LS74
NMI_MAIN
R_NMI
For repair and maintenance
JOTEGO
Sheet: /io/
File: io.kicad_sch
Title: RENEGADE
Size: A4
Date: 2022-07-08
Rev: José Tejada
KiCad E.D.A. 8.0.9
Id: 7/16











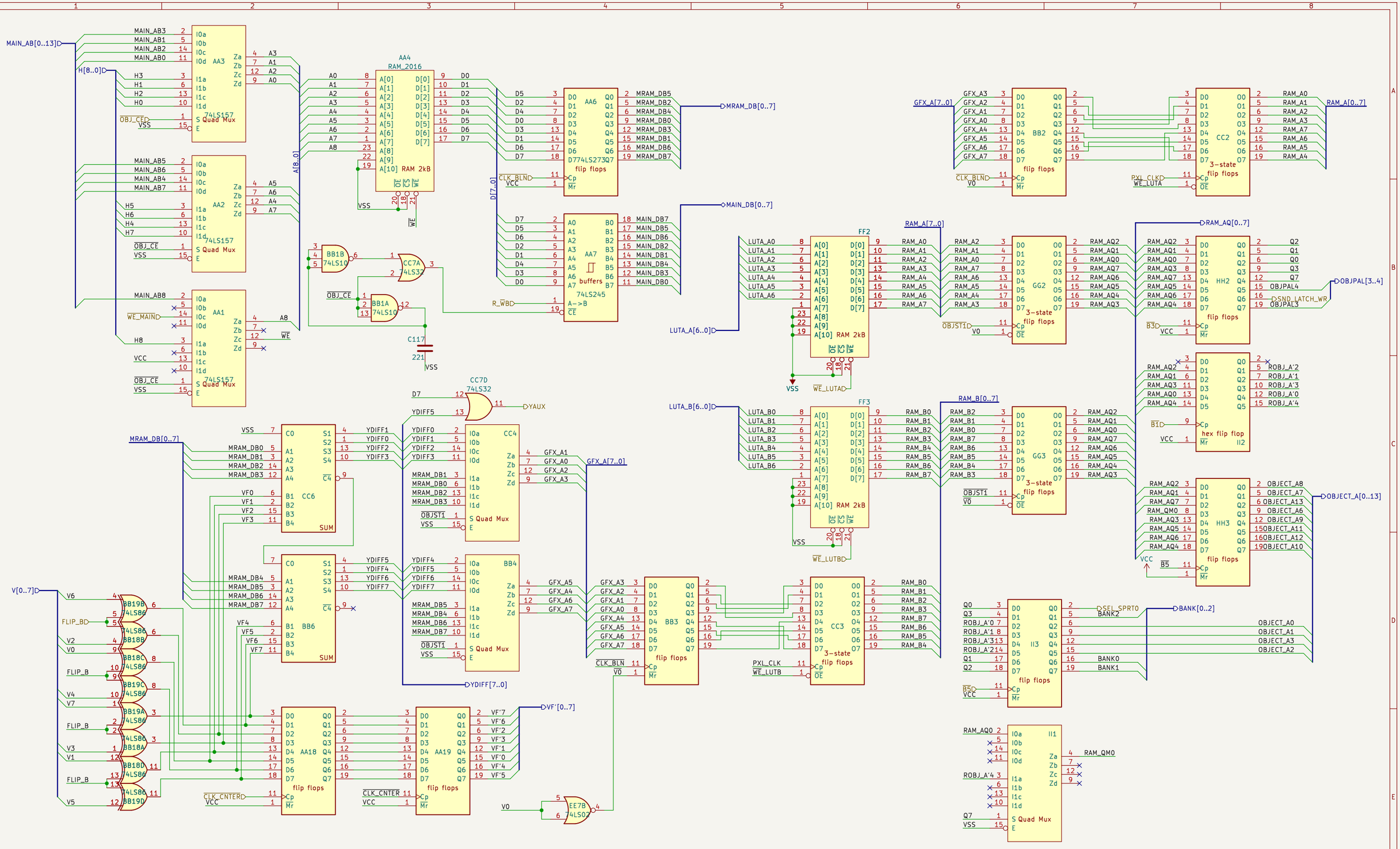
For repair and maintenance

JOTEGO

Sheet: /objects/
File: objects.kicad_sch

Title: RENEGADE

Size: A3	Date: 2022-07-08	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 8/16


MAIN_AB[0..13]
H[8..0]
AA3
AA2
AA1
74LS157
S Quad Mux
AA4
RAM_2016
A[10] RAM 2kB
A[0..8]
D[7..0]
AA6
AA7
74LS273
flip flops
buffers
74LS245
MRAM_DB[0..7]
MAIN_DB[0..7]
R_WB
CLK_BLN
BB1B
BB1A
CC7A
CC7D
74LS32
C117
221
FF2
FF3
LUTA_A[6..0]
LUTA_B[6..0]
WE_LUTAD
WE_LUTBD
RAM_A[7..0]
RAM_B[0..7]
RAM_AQ[0..7]
GFX_A[7..0]
BB2
CC2
3-state
PXL_CLKD
WE_LUTA
GG2
GG3
HH2
hex flip flop
HH3
OBJPAL[3..4]
SND_LATCH_WR
OBJECT_A[0..13]
CC6
SUM
BB4
BB6
CC4
YDIFF[7..0]
YAUX
GFX_A[7..0]
BB19B
BB18B
BB18C
BB19C
BB19A
BB18A
BB18D
BB19D
74LS86
FLIP_BD
FLIP_B
V[0..7]
AA18
AA19
CLK_CNTERD
CLK_CNTER
VF'[0..7]
EE7B
74LS02
BB3
CC3
II3
II2
II1
BANK[0..2]
SEL_SPRT0
BANK2
BANK0
BANK1
RAM_QM0
OBJECT_A0
OBJECT_A1
OBJECT_A2
OBJECT_A3
OBJECT_A7
OBJECT_A9
OBJECT_A10
OBJECT_A11
OBJECT_A12
OBJECT_A13
ROBJ_A'0
ROBJ_A'1
ROBJ_A'2
ROBJ_A'3
ROBJ_A'4
For repair and maintenance
JOTEGO
Sheet: /objects/obj_lut_buf/
File: obj_lut_buf.kicad_sch
Title: RENEGADE
Size: A3
Date: 2022-07-08
KiCad E.D.A. 8.0.9
Rev: José Tejada
Id: 9/16

VERTICAL COUNTER

HORIZONTAL COUNTER

OBJECT STATE MACHINE

SHORTED, BUT NOT CONNECTED

SIGNAL BYPASSED MANUALLY.

REPAIR

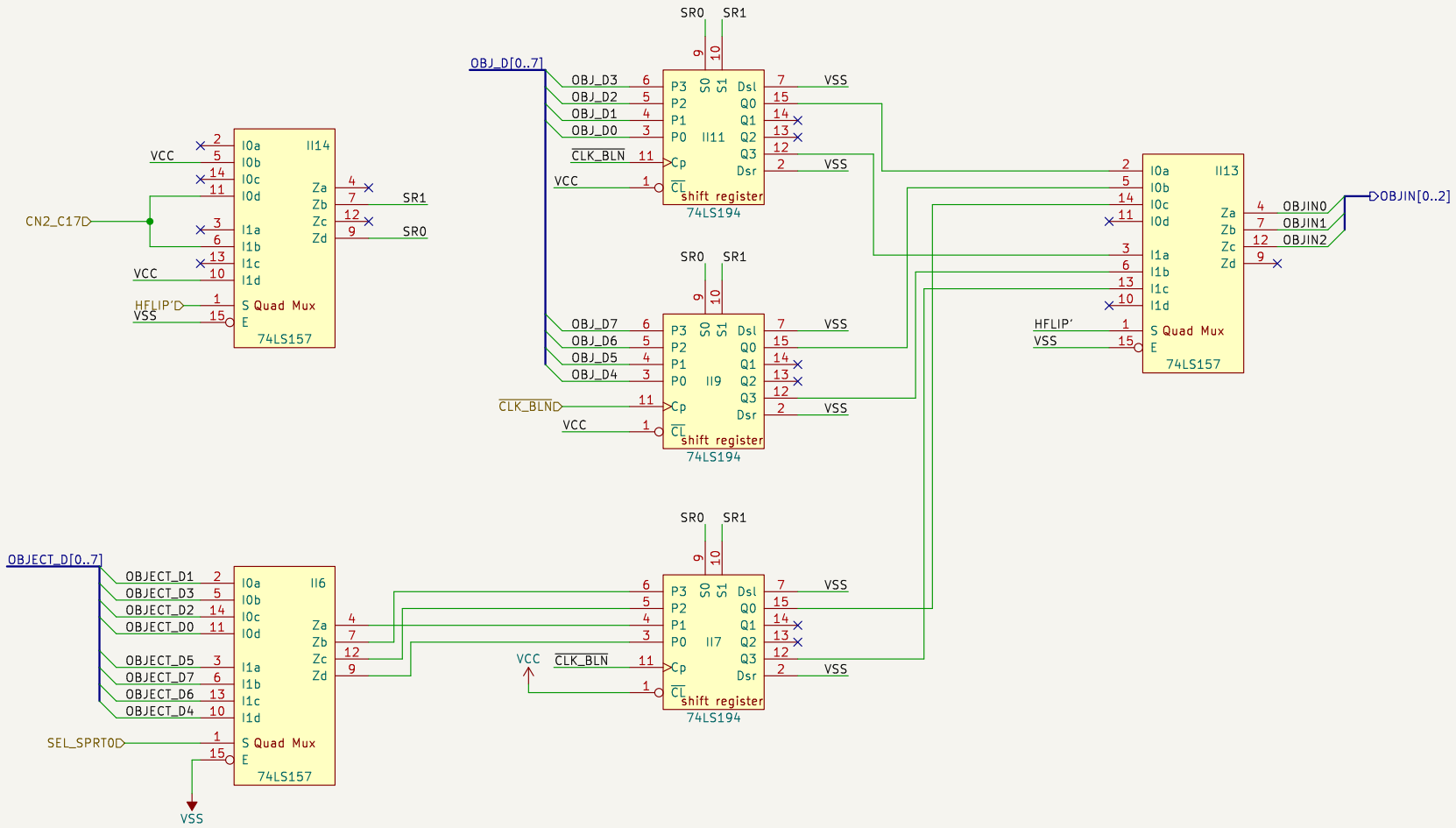
For repair and maintenance

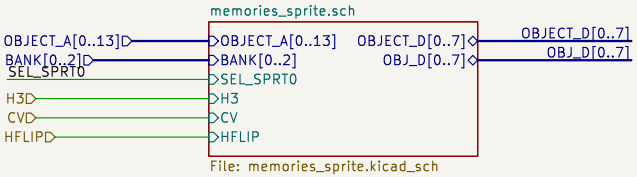
JOTEGO

Sheet: /objects/counters/
File: counters.kicad_sch

Title: RENEGADE

Size: A3 | Date: 2022-07-08 | **Rev: José Tejada**
KiCad E.D.A. 8.0.9 | Id: 10/16





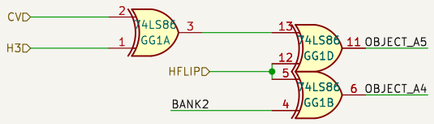
For repair and maintenance

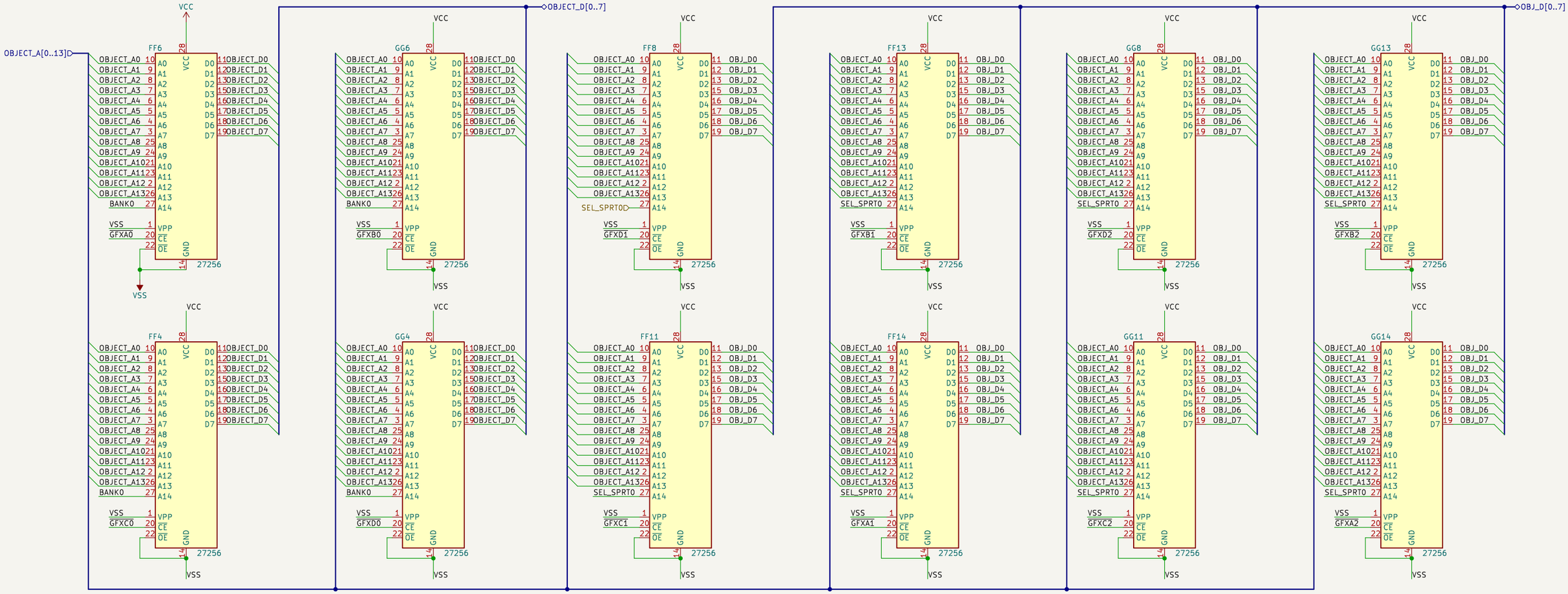
JOTEGO

Sheet: /objects/obj_draw/
File: obj_draw.kicad_sch

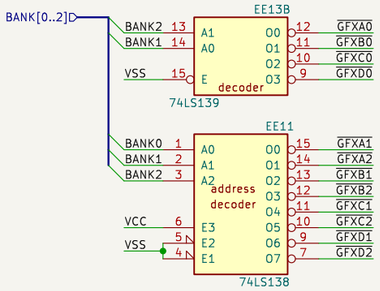
Title: RENEGADE

Size: A4	Date: 2022-07-08	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 11/16





Memorias volcadas



For repair and maintenance

JOTEGO

Sheet: /objects/obj_draw/memories_sprite.sch/
File: memories_sprite.kicad_sch

Title: RENEGADE

Size: A3	Date: 2022-07-08	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 12/16

RAM_AQ[0..7]
OBJBUF[0..4]
OBJIN[0..2]
LINEA[4..0]
LINEB[4..0]
OBJPAL[3..4]
HFLIP
BUFIN3
BUFIN0
BUFIN1
BUFIN2
BUFIN4
LINEA4
LINEA1
LINEA0
LINEA3
LINEA2
LINEB3
LINEB0
LINEB1
LINEB2
LINEB4
OBJBUF4
OBJBUF0
OBJBUF1
OBJBUF2
OBJBUF3
HFLIP'
CLK_H8
REPAIRED
II18 counter 74LS163
II19 counter 74LS163
II15 counter 74LS163
II16 counter 74LS163
II17 S Quad Mux 74LS157
HH17 S Quad Mux 74LS157
EE15 S Quad Mux 74LS157
EE16 S Quad Mux 74LS157
GG19 S Quad Mux 74LS257
FF19 74LS257
FF17 S Quad Mux 74LS157
GG17 S Quad Mux 74LS157
FF15 74LS257
GG15 74LS257
RAM_2016 GG18 A[10] RAM 2kB
RAM_2016 GG16 A[10] RAM 2kB
FF18 hex flip flop 74LS174
FF16 hex flip flop 74LS174
HH1 hex flip flop 74LS174
HH19C HH19D HH19A HH19B 74LS86
HH18C HH18D HH18A HH18B 74LS86
HH16C HH16D HH16A HH16B 74LS86
HH15C HH15D HH15A HH15B 74LS86
EE14B 74LS27
EE13B 74LS04
BB1C 74LS10
CC1B 74LS74
C118 100
C301 101
R54 330Ω REPAIRED
VCC
VSS
WE_CLK
MR_A
MR_B
PE_LINEA
PE_LINEB
PXL_CLK
FLIP_BD
VHQD
H8_QD
B7
LINE
BLANK
BLANK'

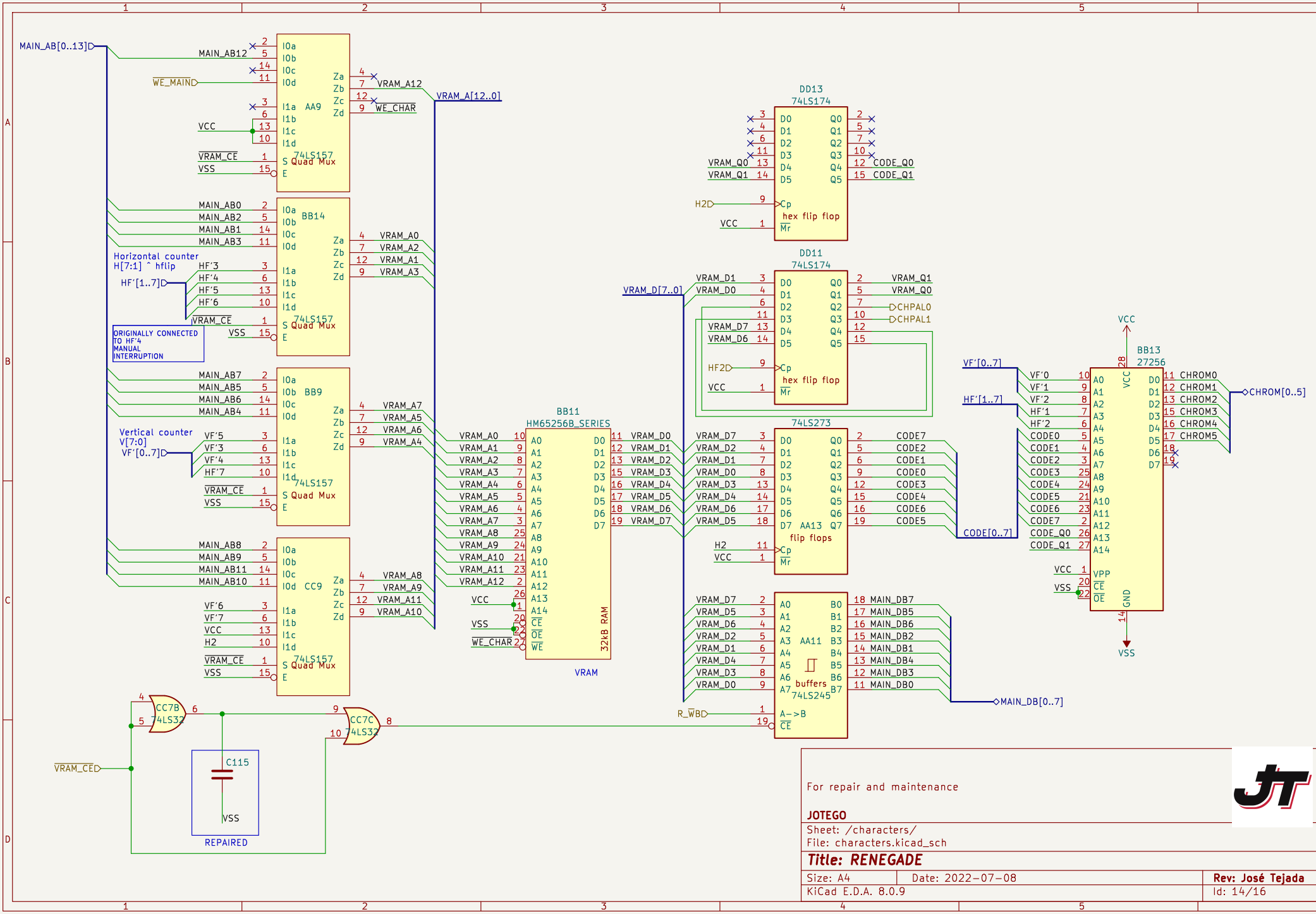
For repair and maintenance

JOTEGO

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File: double_line_buffer.kicad_sch

Title: RENEGADE

Size: A3	Date: 2022-07-08	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 13/16



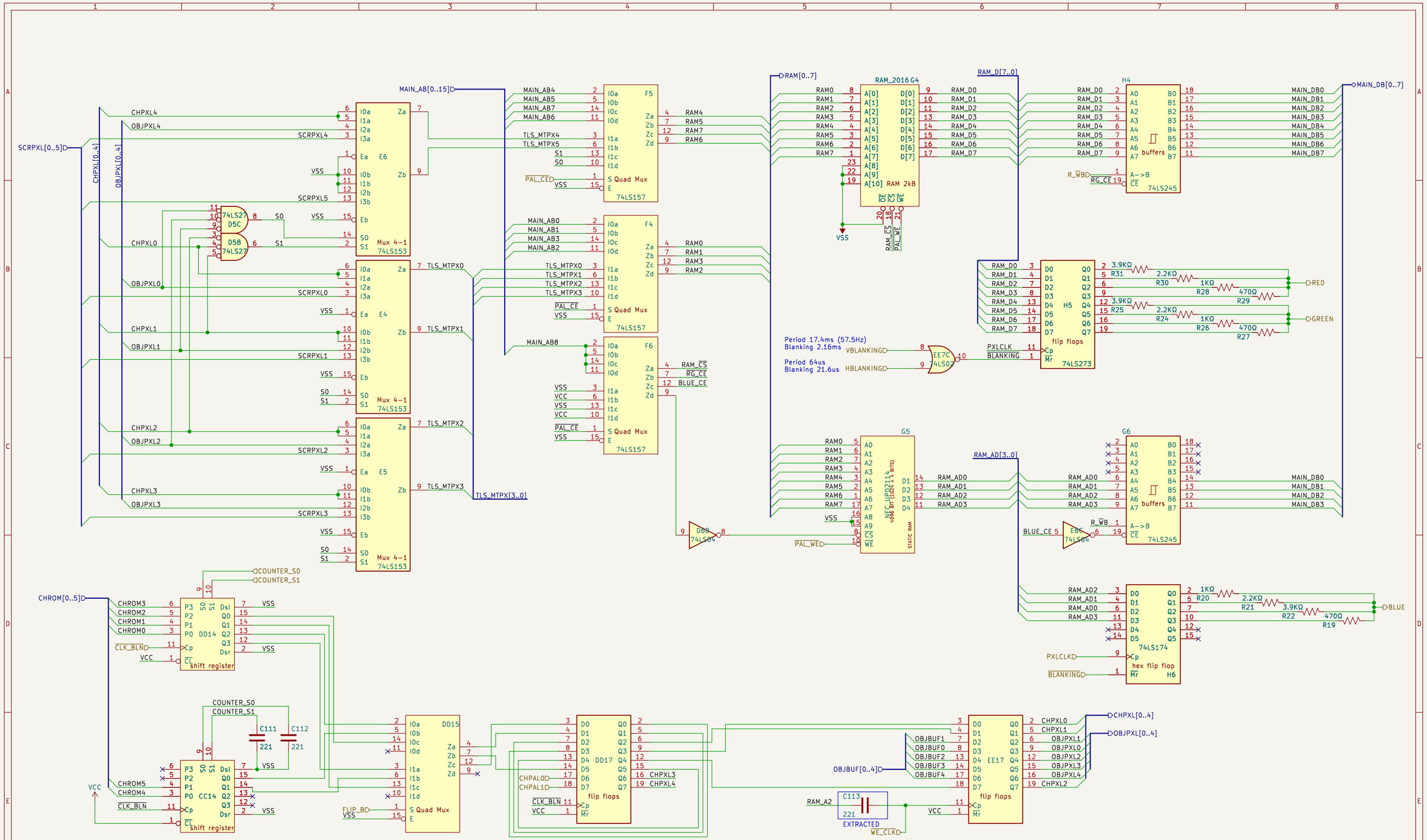
For repair and maintenance

JOTEGO

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Size: A4	Date: 2022-07-08	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 14/16



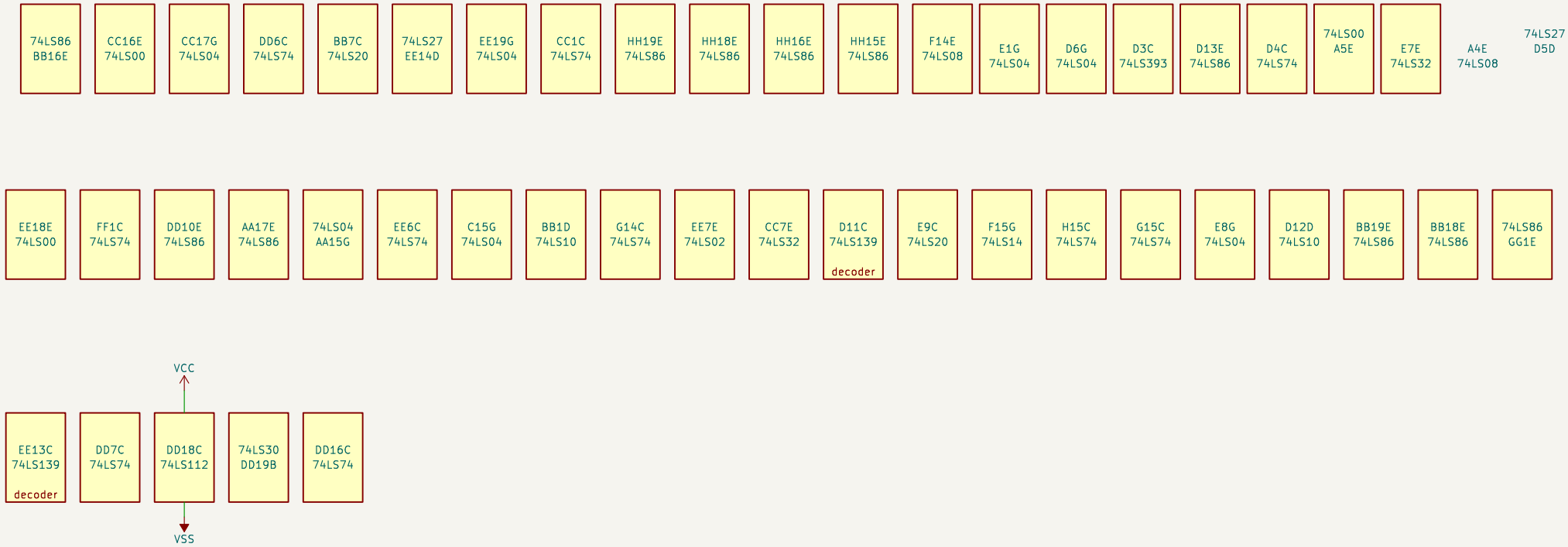
For repair and maintenance

JOTEGO

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Title: RENEGADE

Size: A3	Date: 2022-07-08	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 15/16



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Sheet: /last unit/
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Title: RENEGADE

Size: A4	Date: 2022-07-08	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 16/16